

w2L10 - UCB1:

Regret optimality

- Initial loss exploration - try to minimize

Most popular algorithm - gives not too bad regret bound

- Upper Confidence Bound (UCB algo)

Reference in papers - it will be very different notations in Book

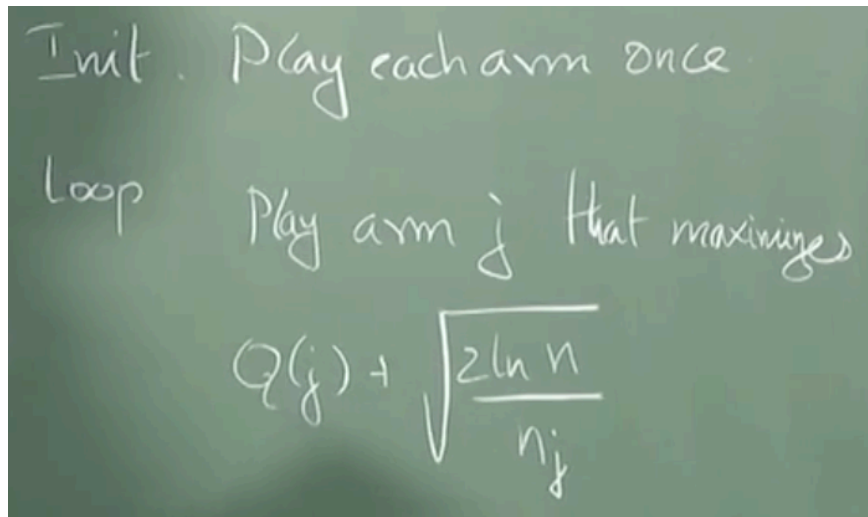
Assuming there are K arms

- Notion of time -> if time is discrete denote by n, if continuous use t
- Assume: There is some prob arbitrary distr don't know - position, bernoulli, gaussian
- Assume: distribution is stationary

Initialization phase:

- Play each arm at least once

Loop:



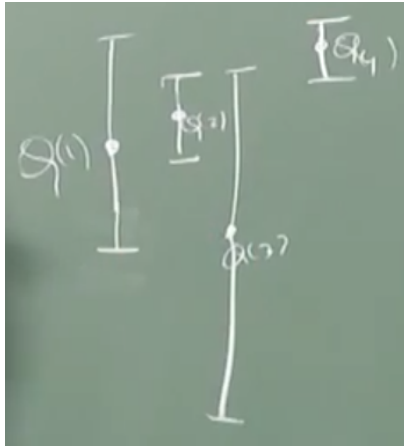
$Q(j)$ - estimated payoff - value fn at time n

Taking current estimating - playing arm which is giving highest value for this expression

$Q(1)$ Q of arm 1, 4 arms

I am not going to use expected estimation so far,

- What is the confidence with which we say that is the expectation
- True value will lie within that band
- More no. of times sampled, it will be narrower band of uncertainty, more confident
 - Q_1 will go to Q^*

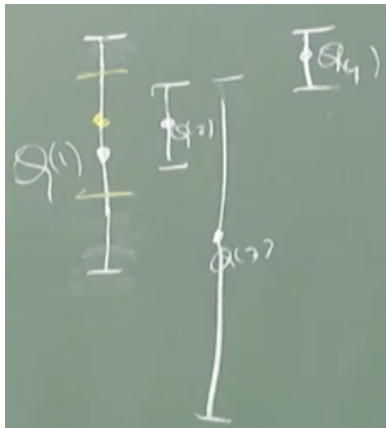


what arm do you taking

Arm 4 could have taken lot of time

Not inclined not instead of arm 3 - chance that it will truly be high is very small

For Arm 1 it may be very high may lie on higher side, it may be higher than Q4



When Q1 shrinks, it may not be attractive

n is number of times arm is played

n_j - no. of times j arm is played

n_j Larger samples, the interval will keep coming down

$$Q(j) + \sqrt{\frac{2 \ln n}{n_j}}$$

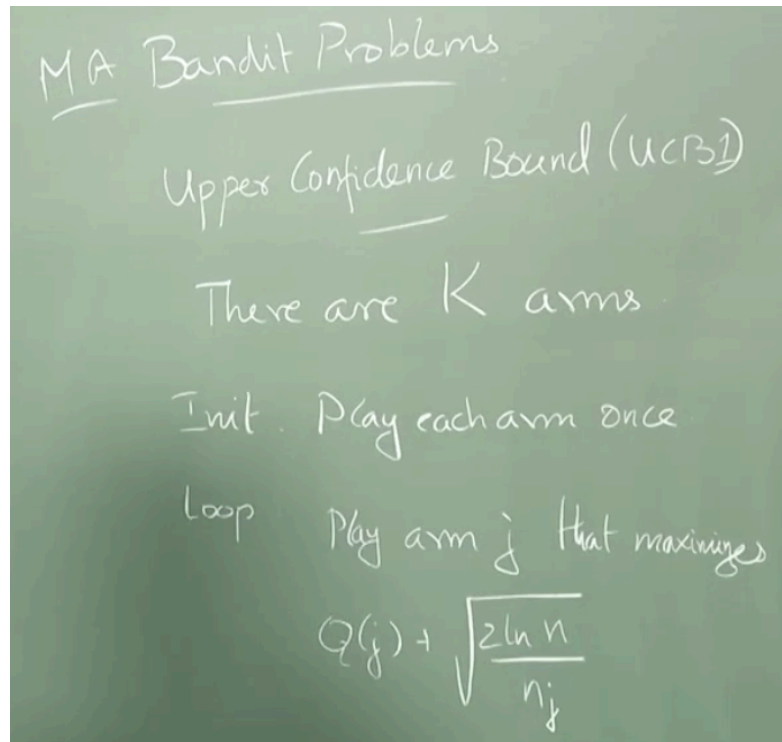
Very simple algo, apart from Q_j , keep track of n_j , in increment you were already doing
(if you are using constant alpha - you may have to store it)

It is deterministic algorithm

$2 \ln n$ will not matter as it will be relative

Rewards are bounded by 1 and 0, you can rescale rewards

w2L11 Concentration Bounds:



UCB without qualifier = UCB1 (Normal UCB < UCB2, UCB improved)

- Each gives slightly better theoretical results
- UCB1 is fairly trivial
- UCB1 has arbitrary reward
- Other variations assume - bernoulli

Thm: For all $K > 1$, if UCB1 is run on K arms having arbitrary reward distr., with support in $[0, 1]$ then its expected regret after any number of plays n is at most

$$8 \sum_{i: q_{i^*}(i) < q_{i^*}(a^*)} \left(\frac{\ln n}{\Delta_i} \right) + \left(1 + \frac{\pi^2}{3} \right) \left(\sum_{j=1}^K \Delta_j \right)$$

$$\Delta_i = q_{i^*}(a^*) - q_{i^*}(i)$$

a^* - optimal action

Regret is upper bounded by : 8 times $\ln$ of non-optimal arms + constant

- Have to exclude optimal arm, not including as it is zero
- Δ_i is zero for non-optimal action

Stationary:

- Whenever i flip the coin prob of head/tail be same

How did we get this expression:

$$Q(j) + \sqrt{\frac{2 \ln n}{n_j}}$$

$T_i(n)$ - no. of times played arm-i in first n trials

$$\text{Regret}_n = \sum_i E[T_i(n)] \Delta_i$$

Formal way of writing regret:

Add regret i times i have played the action-i

$$X_{i,n}$$

rv denotes reward - action-i at time n - what reward i get

Assuming everything is stationary, Expectation of it will be - it will not depend on n

$$E[X_{i,n}] = q_{\pi}(i)$$

To show the result will show that circled term is bounded

$$\text{Regret}_n = \sum_i E[T_i(n)] \Delta_i$$

We show that

$$E[T_j(n)] \leq \frac{8}{\Delta_j^2} \ln n + c$$

constant will give second term

$$\Delta_i = q_{\pi^*}(a^*) - q_{\pi^*}(i)$$

If i played best arm $q^*(a^*)$ - loss for playing arm-i

Sum over all summation

For one of them - for optimal it will be zero

Start from time-0, pull for 100 times,

- Again do it
- Average of various trials (arm in time 3 or 100 prob will be same)

Concentration bounds = concentration inequality = large deviations bounds

Relate true expectations (variance etc. statistics) with

- From some samples - i am estimate
- As no. of samples increases - how quickly/slowly it the expectation approaches true expectation

Chernoff - Hoeffding bound:

Chernoff-Hoeffding bound:

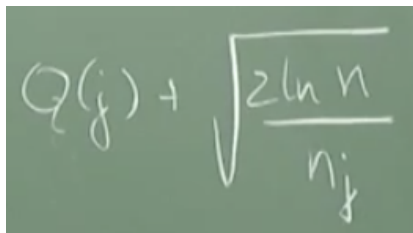
Writing specialised form of bound (it is slightly more general - refer wikipedia/ mathematica)

Let $X_1, X_2, \dots, X_n$ be random variables with common range $[0,1]$ such that

$$E[X_t | X_1, \dots, X_{t-1}] = \mu. \text{ Let } S_n = \frac{X_1 + X_2 + \dots + X_n}{n}.$$

All rewards will have same mean μ (as rewards are stationary)

Q_j is on realisation of S_n random variable


$$Q(j) + \sqrt{\frac{2 \ln n}{n_j}}$$

How can you relate Q_j with Q^*j

This is how it is applicable in our case

$$P(S_n \geq \mu + \epsilon) \leq e^{-2\epsilon^2 n}$$

$$\text{Then } \forall \epsilon \geq 0 \quad P(S_n \leq \mu - \epsilon) \leq e^{-2\epsilon^2 n}$$

Prob that S_n (estimated mean) will be far away from μ

- Smaller the epsilon - larger prob of making error
- For being confidence - for epsilon to be small n should be large

w2 L12 UCB1 Theorem:

Theorem: UCB without clarifier -> UCB1. Many improvements

- Can have arbitrary reward, others assume bernoulli reward (for improving results)

Fairly trivial for Analysis - give flavour of work done

Theorem:

For all $K > 1$, if UCB1 is run on K arms having arbitrary reward distributions with support in $[0, 1]$ then its expected regret after any number of plays n is at most

$$8 \sum_{i: q_*(i) < q_*(a^*)} \left(\frac{\ln n}{\Delta_i} \right) + \left(1 + \frac{\pi^2}{3} \right) \left(\sum_{j=1}^K \Delta_j \right)$$

Where $\Delta_i = q_*(a^*) - q_*(i)$ expected loss you will incur wrt optimal a^*

Essentially this means the loss that you will incur if you play i , the expected loss that you will incur if you play i instead of playing a^* , right you remember what a^* is, the max over a of $q^*(a)$ right so that is a^* , $\operatorname{argmax}_a q^*(a)$ is a^* right so I will always denote by a^* the optimal action right so, what does this tell you, so the regret right is upper bounded by

$$8 \sum_{i: q_*(i) < q_*(a^*)} \left(\frac{\ln n}{\Delta_i} \right)$$

the summation running over all the non-optimal arms, why all the non-optimal arms? Because Δ_i is 0 for optimal arm, so I do not really want that to figure in the summation okay, plus

$$\left(1 + \frac{\pi^2}{3} \right) \left(\sum_{j=1}^K \Delta_j \right)$$

And here I am being little sloppy I have to exclude the optimal arm here also but I do not because it is 0 anyway, Δ_j is 0 anyway so, I mean symmetrically just as I have this summation here I should have the summation there, but I can be a little sloppy and I can write it like that because, I do not have to worry about the additional 0 terms I'm adding okay, great.

$T_i(n)$, that's the term that tells you the number of times it is a random variable that represents the number of times I have played arm i in the first n trials okay.

Write regret after n trials as

$$\text{Regret}_n = \sum_i E[T_i(n)] \Delta_i.$$

Formal definition of regret:

the regret that I expect to have after n time steps is I will take each action, look at the number of times I would have played that action till that time right, multiply it by the regret of playing that action right. So if I played the optimal action then the regret is 0, so if I play some sub optimal action then I will be adding that Δ_i times, the number of times I have played that sub-optimal action i right.

$X_{i,n}$ it is a random variable that denotes the reward that I obtained for playing action i at time n , some n okay some arbitrary n , time n if I play action i right what is the reward

since we are assuming everything is stationary right so the expectation of this will be

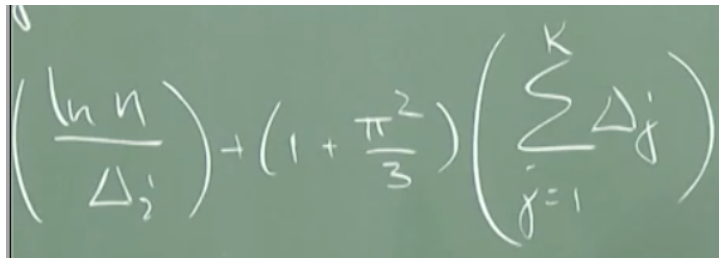
$$E[X_{i,n}] = q_*(i) \quad \text{for stationary it will not depend on } n$$

since we are assuming things are stationary this expectation is really not going to depend on n , so then it becomes expected reward I get for pulling arm i which is essentially $q^*(i)$

$E[T_i(n)]$ guy is bounded. We're going to show that

$$E[T_j(n)] \leq \frac{8}{\Delta_j^2} \ln n + C$$

since i multiply by another Δ_i here to get the regret so that will get me the first term in the regret and some constant here which will get me the second term in the regret.



expected loss you would accrue for playing arm i .

If I had played the best arm over and over again right, if I played the best term over and over again I would get $q^*(a^*)$

if I play arm i , I am going to get $q^*(i)$ right, so the loss I am going to get is $q^*(a^*) - q^*(i)$ correct

I am denoting by Δ_i this is the loss that I expect to accrue for playing arm i , just playing with that one arm right and if I keep playing that arm repeatedly the total loss I am going to accrue is Δ_i times the number of times I will play i right

sum this over all the arms

So one of these arms in the summation or one or more of these arms I mean we do not know one or more of these arms in the summation would be optimal.

And for those terms Δ_i will be 0, so the regret, the contribution to the regret will be 0 and all the other sub optimal arms will contribute Δ_i to the regret.

Stationary: I toss the coin the probability of it coming up heads will be the same whether I toss it the first time or whether I toss it at the tenth time. I pull the arm, the probability of it coming heads will be the same and that is the expectation, right so the expectation will be the same right.

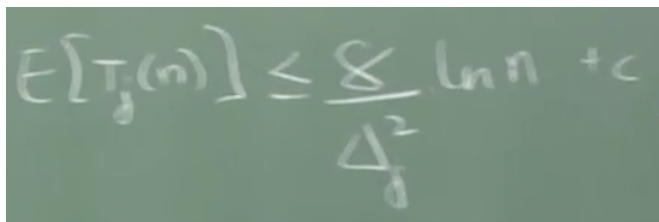
I do several experiments every time starting from time 0 right, I do like millions of experiments okay every time I start from time 0 and I keep pulling arms for some number of time steps

Reset system: reset my whole system start from time 0 and pull for another say 100 time steps something like that right

when I do this experiments like this in the **10th time step** if I pull arm three right so in this million experiments some number of times I would have pulled **arm three** in the **10th time step** okay, then I'll take all the rewards I saw then I'll take the average of that right that is one way of thinking about what this expectation means.

So now you understand right it does not matter whether I pulled arm three at the 10th time step or whether I pull arm three at the thousandth time step if I take this expectation across these million trials, it will be the same.

What we are trying to show is this:


$$E[T_j(n)] \leq \frac{8}{\Delta_j^2} \ln n + c$$

if you are able to show this then we are all done as long as this constant here evaluates to

$$1 + \frac{\pi^2}{3}$$

okay our proof is all done okay.

Now we have to now go and start counting

Need one more fact:

Probability theory called **Concentration Inequalities or Concentration Bounds** (large deviation bounds)

these are essentially results that relate the true expectations of distributions okay, with estimated values of those expectations from samples

different kinds of statistics you have different bounds you have bounds on expectations you have bounds on variance right and so on so forth

as the number of samples increases how quickly or how slowly does this estimation approach my true expectation

W2 L13 PAC Bounds

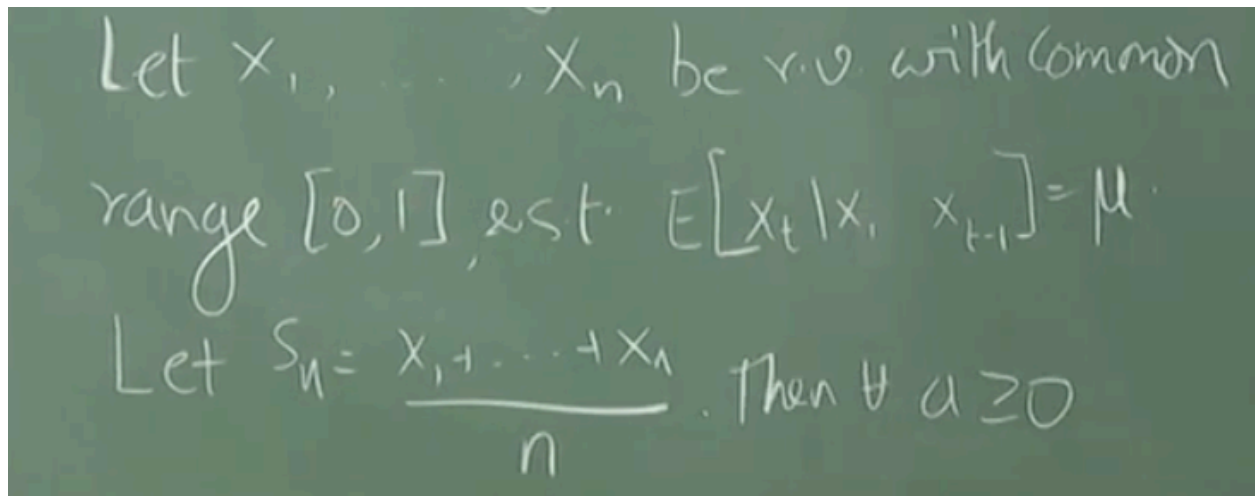
Chernoff bounds or the Chernoff Hoeffding bounds (you should at least know about it):

Chernoff bound is slightly more general than what i am stating

assuming that X_1 to X_n are random variables right each one with a range $[0, 1]$

if it is not in the range $[0, 1]$ you will have to do some kind of normalization in the bound

but for the time being let's just assume it's in the range $[0, 1]$.



expected value of X_t right given all the variables that came before that is μ

all t basically it essentially means that all of the variables have a mean μ

all the variables each one independently X_1 is a random variable with mean μ in the range $[0, 1]$,

X_2 is a random variable with mean μ in the range $[0, 1]$

defining another random variable which we call S_n which is essentially summing up all the X_1 to X_n and divide by n

Chernoff Hoeffding Bound:

Let $X_1, X_2, \dots, X_n$ be random variables with common range $[0, 1]$ such that $E[X_t | X_1, \dots, X_{t-1}] = \mu$.

Let $S_n = \frac{X_1 + X_2 + \dots + X_n}{n}$. Then $\forall \epsilon \geq 0$

as it is stationary reward will be μ , S_n will be Q_j . Q_j is one realisation of random var

Given that taken n samples, how can you relate Q_j with Q^*

$$P(S_n \geq \mu + \epsilon) \leq e^{-2\epsilon^2 n}$$

probability that S_n is greater than or equal to μ plus some ϵ

probability that my S_n will be far away from my μ in one direction, so μ is the true mean, S_n is the estimated mean, the probability that S_n will be greater than at least ϵ from μ right is lesser

If I want ϵ to be very small then n has to be very large only then I will get a high probability

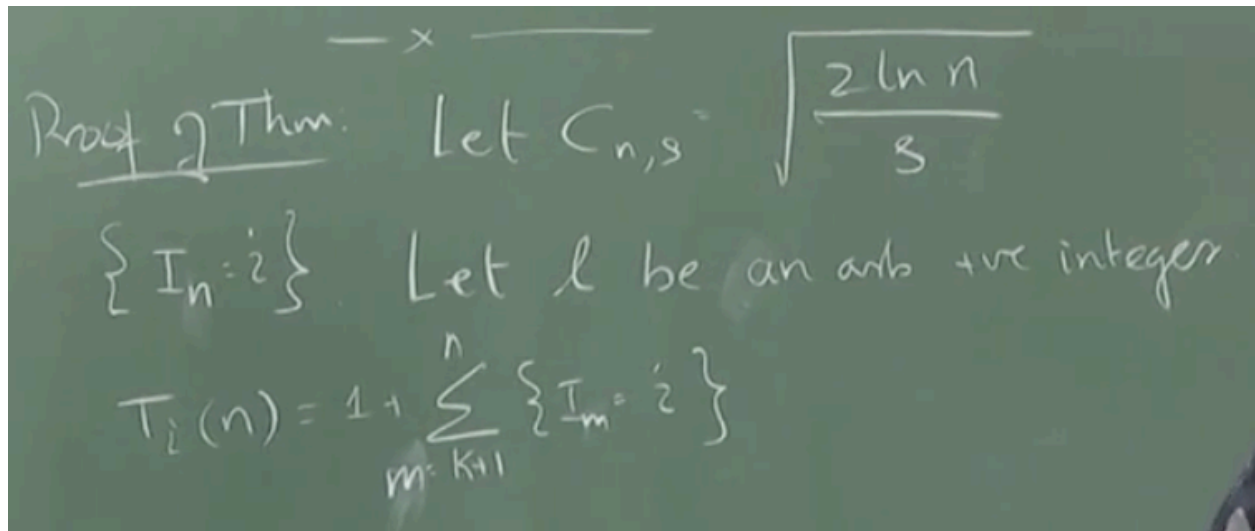
than $e^{-2\epsilon^2 n}$

Smaller the ϵ $\rightarrow$ larger the probability me making an error $\rightarrow$ i need more and more samples n
For high probability if ϵ is small $\rightarrow$ i want n to be large

$$P(S_n \leq \mu - \epsilon) \leq e^{-2\epsilon^2 n}$$

it says in the other direction probability that the estimate I make will be far away from μ , basically below ϵ right is again bounded by the same expression.

7min:



$$T_i(n) = 1 + \sum_{m=K+1}^n \{I_m = i\}$$

You may want to take -AT-TOC (Adv Tech - theory of computer science) course 1/3 course - how to do proof - how to analyse randomised algorithms

After this - bound - taken arm l times

Upto $m-1$ steps l or greater than $l-1$, only then will count arm - should i add

- Look at cumulative count is higher - add it

What we are going to do:

Bound this expression, at least l times have done it l times

What is the value of l

$$\text{Let } C_{n,s} = \sqrt{\frac{2 \ln n}{s}}$$

So instead of writing the cumbersome $\frac{2 \ln n}{s}$ all over right so I am defining that as $C_{n,s}$ okay then if

I write $C_{n-1,s}$ it is $\sqrt{\frac{2 \ln(n-1)}{s}}$ and then I can do $C_{n,s+1}$ so on so forth right I can just like a function

When will i take arm i , when it has highest value for following quantity

$$Q(i) + \sqrt{\frac{2 \ln n}{n_i}}$$

$$\begin{aligned} T_i(n) &= 1 + \sum_{m=K+1}^n \{I_m = i\} \\ &\leq l + \sum_{m=K+1}^n \{I_m = i, T_i(m-1) \geq l\} \\ &\leq l + \sum_{m=K+1}^n \left\{ Q(a^*) + C_{m-1, T_{a^*}(m-1)} \leq Q(i) \right. \\ &\quad \left. + C_{m-1, T_i(m-1)} \right\} \end{aligned}$$

which means sum of Q_i and C term, it should be larger than every other action

- When i will be actually taking action
- Not taking exact number - it will be too involved

Let l be an arbitrary positive integer.

$$\begin{aligned} T_i(n) &= 1 + \sum_{m=K+1}^n \{I_m = i\} \\ &\leq l + \sum_{m=K+1}^n \{I_m = i, T_i(m-1) \geq l\} \end{aligned}$$

$$\leq l + \sum_{m=K+1}^n \{I_m = i, T_i(m-1) \geq l\}$$

$$\leq l + \sum_{m=K+1}^n \{Q_{m-1}(a^*) + C_{m-1, T_{a^*}(m-1)} \leq Q_{m-1}(i) + C_{m-1, T_i(m-1)}, T_i(m-1) \geq l\}$$

Curly braces is no. of times you are counting after action has been taken l times

16min: In above equation. $C_{m-1, t^*a(m-1)} \rightarrow$

- You don't know a^* but it exists

$c(m-1), t(m-1) \rightarrow a^*$ is true true best action

I need at what stage i am talking about q state $\rightarrow q_{m-1} = q_m \rightarrow$ if no action is taken

Need to insert, no. of times i have taken a^* so far, which is $t_{a^*}(m-1)$ in following eq.

$$\leq l + \sum_{m=K+1}^n \left\{ Q_{m-1}(a^*) + C_{m-1, T_{a^*}(m-1)} \leq Q_{m-1}(i) + C_{m-1, T_i(m-1)}, T_i(m-1) \geq l \right\}$$

$$\leq l + \sum_{m=K+1}^n \left\{ \min_{0 \leq s \leq m} \left(Q_{m-1}(a^*) + C_{m-1, T_{a^*}(s)} \right) \leq \max_{l \leq s_i \leq m} \left(Q_{m-1}(i) + C_{m-1, T_i(s_i)} \right) \right\}$$

$$\leq l + \sum_{m=K+1}^n \{Q_{m-1}(a^*) + C_{m-1, T_{a^*}(m-1)} \leq Q_{m-1}(i) + C_{m-1, T_i(m-1)}, T_i(m-1) \geq l\}$$

$$\leq l + \sum_{m=K+1}^n \{\min_{0 < s < m} (Q_s(a^*) + C_{m-1, T_{a^*}(s)}) \leq \max_{l \leq S_i \leq m} (Q_{S_i}(i) + C_{m-1, T_i(S_i)})\}$$

21min: again:

Take min of left side, take max of right

- Look at the number of times one will be higher than the other, then will count as 1
 - The event will be counted, it is overcount of 1st
- In first one, taking about right higher than left at that particular instance m-1

Why writing overcount, it is easier to write expression:

$$\leq l + \sum_{m=1}^{\infty} \sum_{s=1}^{m-1} \sum_{S_i=l}^{m-1} \{Q_s(a^*) + C_{m, T_{a^*}(s)} \leq Q_{S_i}(i) + C_{m, T_i(S_i)}\}$$

First l times, I am not worried, neat trick got rid of min, max

So what are we doing here so assume that the Min occurs at some index right let us call that index, I do not know p, p1

let us assume that the Max occurs at some index say p2 right, so now what I am doing here is I am summing over all possible indices for a-star and

I am summing over all possible indices for i, correct.

So the combination p1, p2 will occur somewhere because I am summing over all possible indices, right.

p1 for s - min was achieved , p2 for sa for max

I will be certianlity be overcounting - still maintaing upper bound

From strict equality (1st eq below), we loosened the bound

$$\begin{aligned}
 T_i(n) &= 1 + \sum_{m=k+1}^n \{I_m = 2\} \\
 &\leq \ell + \sum_{m=k+1}^n \{I_m = 2, \underbrace{T_i(m-1) \geq \ell}\} \\
 &\leq \ell + \sum_{m=k+1}^n \{Q(a^*) + C_{m-1, T_{a^*}(m-1)} \leq Q_{m-1}(\cdot) \\
 &\quad + C_{m-1, T_i(m-1)}, \\
 &\quad T_i(m-1) \geq \ell\}
 \end{aligned}$$

Essentially counting larger and larger event
We can bound following eq.

$$\leq \ell + \sum_{m=1}^{\infty} \sum_{s=1}^{m-1} \sum_{s_i=\ell}^{m-1} \{Q_s(a^*) + C_{m, T_{a^*}(s)} \leq Q_{s_i}(i) + C_{m, T_i(s_i)}\}$$

When can this happen - true if conditions are met:

This event can be true if one of several conditions hold,

$$1) Q_s(a^*) \leq q_*(a^*) - C_{m, T_{a^*}(s)}$$

$$Q_s(a^*) \leq q_*(a^*) - C_{m, T_{a^*}(s)}$$

Either grossly estimating + confidence bound $\leq$ true values

You are grossly over estimating value

$$2) Q_{s_i}(i) \geq q_*(i) + C_{m, T_i(s_i)}$$

$$Q_{s_i}(i) \geq q_*(i) + C_{m, T_i(s_i)}$$

If values are close - don't have to grossly underestimate

$$3) q_*(a^*) < q_*(i) + 2C_{m, T_i(s_i)}$$

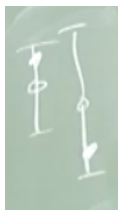
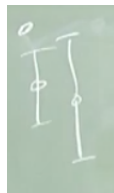
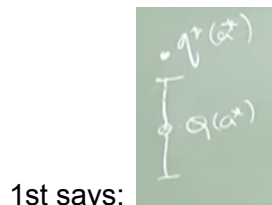
$$q_*(a^*) < q_*(i) + 2C_{m, T_i(s_i)}$$

Add twice confidence interval - from lower value - optimal upper value lies, these two are close together,
 3rd - both don't hold, still i may be making mistake



current uncertainty i have

33min: One of these 3 have to be satisfied



3rd condition none of these need to happen, given current uncertainty in q_i true means are not far apart, they are within uncertainty

Have this eq. In notes:

$$8 \sum_{i: q(i) < q_h(\alpha^*)} \left(\frac{\ln n}{\Delta_i} \right) + \left(1 + \frac{\pi^2}{3} \right) \left(\sum_{j=1}^K \Delta_j \right)$$

37min: Okay, so let us look at 1, so I want to look at the probability that a^*

$$P(Q_s(a^*) \leq q_*(a^*) - C_{m,T_{a^*}(s)})$$

We want to bound the probability. S_n is Q . μ is q^*

Bound becomes right term

Plug n here:

Substitute and cancel things

What is n ?

so

will get $\ln m$

$$P(Q_s(a^*) \leq q_*(a^*) - C_{m,T_{a^*}(s)}) \leq m^{-4}$$

2nd term:

$$P(Q_{s_i}(i) \geq q_*(i) + C_{m,T_i(s_i)}) \leq m^{-4}$$

3rd term:

$$\begin{aligned}
 Q(a^*) &\leq q_{*}(a^*) - C_{m, T_{a^*}}(s) & - (1) \\
 Q_{s_i}(i) &\geq q_{*}(i) + C_{m, T_i}(s_i) & - (2) \\
 q_{*}(a^*) &< q_{*}(i) + 2 C_{m, T_i}(s_i) & - (3)
 \end{aligned}$$

here we will use mysterious l - after that many times taken arm i . If taken sufficiently, this event will not happen

For $l = \left\lceil \frac{8 \ln(n)}{\Delta_i^2} \right\rceil$, event 3 is False!!

$$q^*(a^*) - q^*(i) - 2C_{m, T_i}(s_i)$$

essentially I am saying that my $T_i(s_i)$ is at least this much the term $\left\lceil \frac{8 \ln(n)}{\Delta_i^2} \right\rceil$

$$\begin{aligned}
 (3) \text{ For } l = \left\lceil 8 \ln(n) / \Delta_i^2 \right\rceil, (3) \text{ is false!} \\
 q_{*}(a^*) - q_{*}(i) - 2 C_{m, T_i}(s_i)
 \end{aligned}$$

$$\leq l + \sum_{m=1}^{\infty} \sum_{s=1}^{m-1} \sum_{s_i=l}^{m-1} \{Q_s(a^*) + C_{m,T_{a^*}(s)}\} \leq$$

s_i starts from l

Proof 2 Thm. Let $C_{n,s} = \sqrt{\frac{2 \ln n}{s}}$

Plug in this eq.

Will get Δ_i

So if my s is this much, at least this much, so what do I get, I get $2/8 \ln m$ and my Δ_i will go up so $\ln m$, $\ln m$ will get cancelled, right so $2/8$ will become $1/4$ that is half, half into Δ_i right, so half and 2 will get cancelled I will essentially get Δ_i , right so this expression will become.

$$q_*(a^*) - q_*(i) - 2C_{m,T_i(s_i)}$$

$$= q_*(a^*) - q_*(i) - \Delta_i$$

46min Repeating:

$$q_k(a^*) - q_k(i) - 2C_{m,T_i(s_i)} \\ = q_k(a^*) - q_k(i) - \Delta_i = 0$$

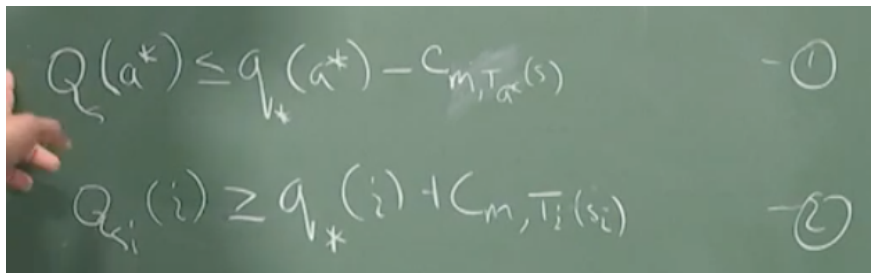
if taken action atleast l times, prob 3rd

cond = 0

Page 20: so the probability of this whole thing happening is upper bounded by just two times this probability

$$E[T_i(n)] \leq \left\lceil \frac{8 \ln(n)}{\Delta_i^2} \right\rceil + \sum_{m=1}^{\infty} \sum_{s=1}^{m-1} \sum_{S_i = \left\lceil \frac{8 \ln(n)}{\Delta_i^2} \right\rceil}^{m-1} \{Q_s(a^*) + C_{m,T_{a^*}(s)} \leq Q_{S_i}(i) + C_{m,T_i(S_i)}\}$$

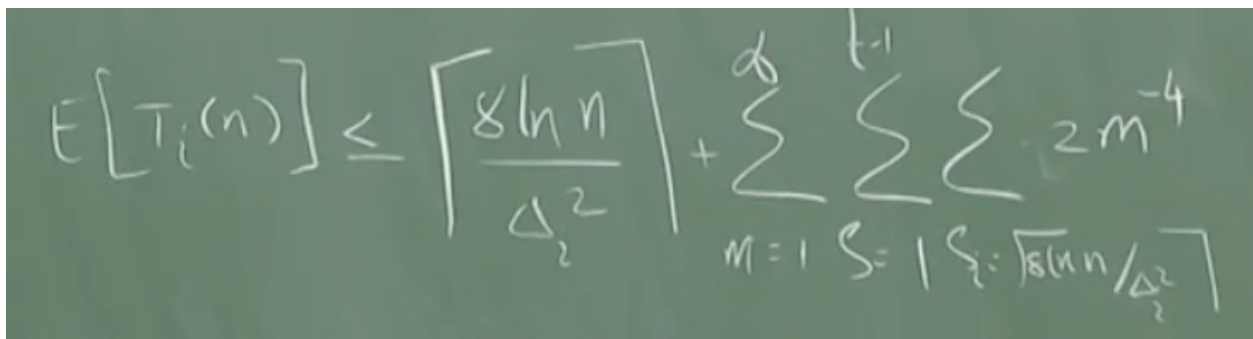
Prob of 1st and 2nd event in following:



$$Q(a^*) \leq q_*(a^*) - C_{m,T_{a^*}(s)} \quad - (1)$$

$$Q_{S_i}(i) \geq q_*(i) + C_{m,T_i(S_i)} \quad - (2)$$

Upper bounded by probability =1, will simplify to below



$$E[T_i(n)] \leq \left\lceil \frac{8 \ln n}{\Delta_i^2} \right\rceil + \sum_{m=1}^{\infty} \sum_{s=1}^{m-1} \sum_{S_i = \left\lceil \frac{8 \ln n}{\Delta_i^2} \right\rceil}^{m-1} 2m^{-4}$$

$$1 + \frac{\pi^2}{3}$$

This summation - declare to converge to

$$\leq \frac{8 \ln n}{\Delta_i^2} + 1 + \frac{\pi^2}{3}$$

simple algebra, will converge to limit

Whatever it converges to some small number

$$E[T_i(n)] \leq \left\lceil \frac{8 \ln n}{\Delta_i^2} \right\rceil + \sum_{m=1}^{\infty} \sum_{s=1}^{t-1} \sum_{\substack{S: |S| = \left\lceil \frac{8 \ln n}{\Delta_i^2} \right\rceil}} 2m^{-4}$$

$$\leq \frac{8 \ln n}{\Delta_i^2} + 1 + \frac{\pi^2}{3}$$

$$\frac{8 \ln(n)}{\Delta_i^2}$$

What is interesting is

This is for one sub-optimal arm, no. of times you have played
Earlier eq.

$$8 \sum_{\substack{i: q_i(z) < \\ q_i(a^*)}} \left(\frac{\ln n}{\Delta_i^2} \right) + \left(1 + \frac{\pi^2}{3} \right) \left(\sum_{j=1}^K \Delta_j \right)$$

Regret grows as $\ln n \rightarrow$ can't have an algo which is better than $\ln n$
If arms are close, it is harder to figure out which are bad

People fiddle around 8

w2 L13 PAC Explanation and Naive Algo Proff (PAC Bounds)

first one was asymptotic correctness and the second one we talked about was regret optimality which we already looked at one example of a regret optimal algorithm UCB

3 PAC optimality:

Chernoff Hoeffding bound, we saw that already.

Chernoff – Hoeffding Bound: Let $X_1, X_2, X_3, \dots, X_n$ be random variables with common range $[0, 1]$ such that $E[X_t | X_1, X_2, \dots, X_{t-1}] = \mu$.

Let $S_n = \frac{X_1 + X_2 + \dots + X_n}{n}$. Then for $\epsilon > 0$,

$$P[S_n \geq \mu + \epsilon] \leq e^{-2\epsilon^2 n}$$

$$P[S_n \leq \mu - \epsilon] \leq e^{-2\epsilon^2 n}$$

Markov Inequality:

$$Pr(X \geq a) \leq \frac{E[X]}{a}; a > 0$$

assuming a is positive, it is prob

Union Bound:

$$P(\cup_i A_i) \leq \sum_i P(A_i)$$

PAC bounds for multi-armed bandit problems

an input which is I will give you an ϵ and a δ and what I am asking for is a guarantee that at the end of this algorithm the arm that I return to you will be ϵ close to the optimal arm with probability $1 - \delta$.

PAC algos take ϵ and δ

PAC Bounds for MAB

Naive Algorithm:

Input: $\epsilon > 0, \delta > 0$

Output: 'An arm'

For each $a \in A$, do

Sample 'a' l times where $l = \frac{2}{\epsilon^2} \ln \left(\frac{2K}{\delta} \right)$

Let $Q(a)$ be the average reward of arm a

End

Output $a = \operatorname{argmax}_{a \in A} Q(a)$

Ref: Paper way way better written - action elimination for RL

median elimination algorithm by Eyal Even-Dar, Shie Mannor and Yishay Mansour okay, it is titled Action elimination for Reinforcement Learning okay so the nice thing about this ok I will tell you the nice things about the paper later.

k = no. of arms

Take each arm - pull it many times

Until got good estimate of expectation

Do for each arm - once i have good estimate

Use that arm

How did i produce magic number

ϵ - δ PAC algorithm:

Theorem 1: Naive (ϵ, δ) is an (ϵ, δ) - PAC

$$O \left(\left(\frac{K}{\epsilon^2} \right) \log \left(\frac{K}{\delta} \right) \right).$$

With arm sample complexity (no. of times you pull arm)

Interesting to show it is actually ϵ - δ PAC. Proof ->

Proof:

Let a' be an arm such that $q_*(a') < q_*(a^*) - \epsilon$.

$$\begin{aligned} P(Q(a') > Q(a^*)) &\leq P(Q(a') > q_*(a') + \epsilon/2 \text{ OR } Q(a^*) < q_*(a^*) - \epsilon/2) \\ &\leq P(Q(a') > q_*(a') + \epsilon/2) + P(Q(a^*) < q_*(a^*) - \epsilon/2) \\ &\leq e^{-2\frac{\epsilon^2}{4}l} * 2 \end{aligned}$$

$$Q(a') > Q(a^*)$$

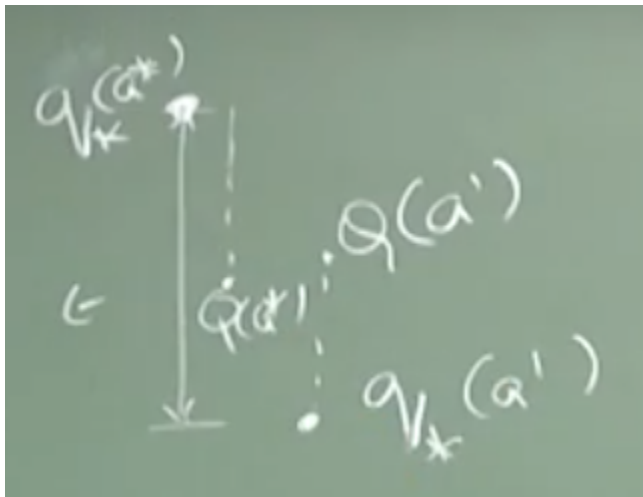
so look at the prob and bound it

Proof

Let a' be an arm s.t. $q_*(a') < q_*(a^*) - \epsilon$

$$P(Q(a') > Q(a^*)) \leq$$

$$P(Q(a') > Q(a^*)) \leq P(Q(a') > q_*(a') + \epsilon/2 \text{ OR } Q(a^*) < q_*(a^*) - \epsilon/2)$$



18min? this is the case I am looking at q_{a^*} more than 2 times above or below? It has to be ep away. Over estimate or under estimate

Taking independent events and adding (making things simpler) - below

$$P(Q(a') > Q(a^*)) \leq P(Q(a') > q_{*}(a') + \epsilon/2 \text{ OR } Q(a^*) < q_{*}(a^*) - \epsilon/2) \\ \leq P(Q(a') > q_{*}(a') + \epsilon/2) + P(Q(a^*) < q_{*}(a^*) - \epsilon/2)$$

Can apply chernoff bound:
Two times bound

$$\leq P(Q(a') > q_{*}(a') + \epsilon/2) + P(Q(a^*) < q_{*}(a^*) - \epsilon/2) \\ \leq 2e^{-2 \frac{\epsilon^2}{4} \ell}$$

Plug in ℓ there

foreach $a \in A$ do.
Sample a ℓ times.
 $\ell = \frac{4}{\epsilon^2} \cdot \ln\left(\frac{2K}{\delta}\right)$.
Let $Q(a)$ be the av. reward.

foreach $a \in A$ do.
Sample a ℓ times.
 $\ell = \frac{4}{\epsilon^2} \cdot \ln\left(\frac{2K}{\delta}\right)$.
Let $Q(a)$ be the av. reward.

Adjust ℓ so things get canceled out

What do you think will be left of - what will be the prob me choosing one non-optimal arm by

$$\frac{\delta}{K}$$

mistake
Plug here

for each $a \in A$ do.
 Sample a l times.
 $l = \frac{2}{\epsilon^2} \ln\left(\frac{2K}{\delta}\right)$

Will get δ / K

How we came up with magic number el

$$\begin{aligned}
 P(Q(a) > Q(a^*)) &\leq P(Q(a) > q_{\star}(a) + \epsilon/2 \text{ OR } Q(a^*) < q_{\star}(a^*) - \epsilon/2) \\
 &\leq P(Q(a) > q_{\star}(a) + \epsilon/2) + P(Q(a^*) < q_{\star}(a^*) - \epsilon/2) \\
 &\leq 2e^{-2 \frac{\epsilon^2}{4} l} \\
 \text{Subst. } l, \quad P(Q(a) > Q(a^*)) &\leq \delta/K
 \end{aligned}$$

Above is for one a^*

$$\text{Summing over all } a', \\ \text{Prob of 'failure'} \leq (K-1) \cdot \frac{\delta}{K} \leq \delta.$$

Naive algo: analysis is straightforward -

- If give algo, you can derive

Thm 1: Naive(ϵ, δ) is an(ϵ, δ)-PAC
algo, with arm sample complexity
 $O\left(\left(\frac{K}{\epsilon^2}\right) \cdot \log\left(\frac{K}{\delta}\right)\right).$

it is impossible to get rid of $1/\epsilon^2$ and $\log(1/\delta)$

When i come up with other PAC algorithm

- Achieve with as few samples as possible
- Sample complexity is at these 2 places

Can you get rid of k ? You have to try atleast one so can't get rid of first K

Can try to get rid of $\log(k)$

Once i reach there, get rid of 2

$$t = \frac{2}{\epsilon^2} \cdot \ln\left(\frac{2K}{\delta}\right)$$

or put 1.77

So I will correct it to K right okay so how do we prove it well the sample complexity part is obvious we do not have to prove that we just have to prove the (ϵ, δ) PAC so for that we will assume that let a' be an arm such that $q^*(a')$ okay so a' is an arm which is not ϵ optimal right, an arm that is ϵ optimal is ϵ close to the best arm right so $q^*(a^*)$ is the reward corresponding to the best arm we know that right so I am saying that I will pick an arm a' such that it is at least ϵ away from $q^*(a^*)$ right.

So the $q^*(a')$ is at least ϵ away from $q^*(a^*)$ right so that means it is a bad arm right so arms that are more than ϵ close to the true arm I do not care right if I select them I am happy because my guarantee is satisfied I really have to worry about making a mistake which in this case is selecting an arm that is more than ϵ away from the true optimal okay so basically we are looking at the event right so this is what we want we are looking at right so if this is the case, ie. $Q(a') > Q(a^*)$ then the algorithm will output a' instead of a^* correct.

So people can see why we need this condition right so if a' has to be reported as the best arm it has to be at least better than a^* right great so that is the condition so already we are being little loose here okay, this is not really the probability that yes a' will be the best arm okay but it is only looking at some sub event of that it is not all the events together but this will give me at least a bound on the probability so that is essentially why you are looking at this event okay.

Right so what is happening here, let us say that is my that is my $q^*(a^*)$ okay that is my right and this gap is at least this gap is at least ϵ , okay so there is this is the setup that we have now so and I am making some estimate of $q^*(a')$ which is $Q(a')$ right and I am making some estimate of $q^*(a^*)$ which is $Q(a^*)$ so the condition I am asking is that $Q(a')$ should be higher than I mean $Q(a^*)$ should be higher than $Q(a^*)$ right.

So essentially I am saying that I should have $Q(a')$ here $Q(a^*)$ somewhere lower right so this is the case I am looking at right so this all automatically means that either $Q(a')$ has to be more than $\epsilon/2$

from $q^*(a')$ right it should be at least ϵ more than $\epsilon/2$ above or $Q(a^*)$ should be more than $\epsilon/2$ below $q^*(a^*)$, right if they are exactly $\epsilon/2$ they will be equal I mean they won't be one above the other right if one is $\epsilon/2$ below other is $\epsilon/2$ above so you can think of cases where this is actually here right so this could be here in which case q^* I mean $q^*(a^*)$ has to be even below right it will be more than ϵ away forget about $\epsilon/2$ right so likewise for cases where you are overestimating $Q(a^*)$

Then this will also have to be higher than that in which case again it will be more than ϵ away right so the interesting case is when it is in between the two right then it has to at least one of them has to be $\epsilon/2$ away from the true estimate right either as an overestimate or as an underestimate right is it clear so why I wrote this down, right one of these two events has to be satisfied. right I am just taking them as independent events and summing them up and that still be an upper bound on the total probability you have right, making things simpler and simpler for me and now what happens here it kind of in the Chernoff space right.

So I can apply Chernoff bound here so what would be the case here $e^{-2\epsilon^2/4}$, yeah so let us do that so this will be $\leq e^{-2\epsilon^2/4}$, what about the other term, the same right it is a symmetrical bound so if that is $\epsilon/2$ or $-\epsilon/2$, it is going to be the same right so is it clear okay so now plug in this I

there so what do you get, hmm, so essentially what we need to do now is adjust this l so that all those things get cancelled out that is basically what we are looking for right and you will be left with, what should you be left with.

What should you be left with what do you think you should be left with no no no I want this probability δ this is the probability of me making a mistake right so it should be δ , should I be left with δ , not quite this is the probability of me choosing one non optimal arm by mistake what if all my arms are non-optimal except the one optimal arm right everything is away from that is the worst case so I should have δ/K or $K-1$ so we will take δ/K , okay that is easier than and so this whole thing should simplify to δ/K right so that is what I should be looking for so what should be my l right.

So this if we plug this in what will I get so the $4\epsilon^2$ will go so I will get $2 \ln(2K\delta)$, that is not quite right and what do I get then, yeah 2 into $2K\delta$ so I will get δK , I will get δK is it fine I simplify this I get δK so now you know how you actually come up with this magic number I right, by actually deriving this bound and then figuring out what else should I plug in here so that I get the necessary error for the probability okay so that is just for one a' hmm there is a 2 outside right, 2ϵ okay.

So this is clear, this is a naive algorithm and the analysis also is pretty simple, pretty straightforward right so all of you are happy with this right so if you if I give you an algorithm with a slightly different way of selecting actions you can all derive the complexity the l that is required basically great it turns out that so we have produced this complexity here right it turns out that it is almost impossible to get rid of this $1\epsilon^2$ right and $\log(1/\delta)$ right, so when I try to come up with another PAC algorithm right I want to you remember what I said about PAC analysis.

I have to try every arm at least once so I cannot get rid of the K also right so the only thing I can try and minimize here is well I can try and get rid of this K right so I can try to get rid of the $\log K$ dependency right and then have $K\epsilon^2 \log(1/\delta)$ roughly I can try to move in that direction and once I reach there I will probably be in, left with trying to get rid of this two here right and try to make the 2 into $3/2$ or 1.77 or something like that. you know how these things proceed right so you can try to optimize the constants but more or less this $1\epsilon^2$ and $\log(1/\delta)$ is something that will have to live with okay.

$$\text{Plug in } l = \frac{2}{\epsilon^2} \ln\left(\frac{2K}{\delta}\right)$$

$$\begin{aligned}
P(Q(a') > Q(a^*)) &\leq 2e^{-\frac{1}{2}\epsilon^2 \times \frac{2}{\epsilon^2} \ln\left(\frac{2K}{\delta}\right)} \\
&\leq 2e^{-\ln\left(\frac{2K}{\delta}\right)} \\
&\leq 2 \times \left(\frac{2K^{-1}}{\delta}\right) \\
&\leq \frac{\delta}{K}
\end{aligned}$$

So that's just for 1 a' . Now summing over all a' ,

$$\text{Prob. Of failure} \leq (K - 1) \frac{\delta}{K} \leq \delta$$

So let us make things a little formal so we call this algorithm naive, so then algorithm Naive which takes as input ϵ, δ is an (ϵ, δ) PAC algorithm, you remember what is (ϵ, δ) PAC, we defined it in the earlier lecture right (ϵ, δ) PAC gives you that whatever guarantees you are talking about right (ϵ, δ) PAC algorithm with arm sample complexity essentially the number of times you pull the

arms given by $O\left(\left(\frac{K}{\epsilon^2}\right) \log\left(\frac{K}{\delta}\right)\right)$ right so this part is obvious right I pull each arm $\frac{1}{\epsilon^2} \ln \frac{K}{\delta}$ right so for n arms I have to do that n times so my overall sample complexity is $O\left(\left(\frac{K}{\epsilon^2}\right) \log \frac{K}{\delta}\right)$ okay, so the interesting part now is to show that it is actually (ϵ, δ) PAC, what is it obviously yeah for today's lecture if I write n by mistake right please point it out to me.

w2 L14 PAC bound for MAB

Markov Inequality:

$$Pr(X \geq a) \leq \frac{E[X]}{a}; a > 0$$

Union Bound:

$$P(\cup_i A_i) \leq \sum_i P(A_i)$$

Upper bounding joint event by individual events

Okay so we talked about multiple solution concepts for bandit algorithms right so the

- first one was asymptotic correctness and
- the second one we talked about was regret optimality which we already looked at one example of a regret optimal algorithm UCB or UCB1 more correctly even though popularly when people say UCB they refer to UCB1 right and
- the third one PAC optimality right which was the probably approximately correct thing where you are trying to bound with some probability right that I mean you are close enough to the true solution right

Simple Algorithm: So that is essentially the guarantee that we want to give right.

Input: epsilon and delta

- Asking for guarantee
- Arm which it will return - with high probability (1-delta) - will give arm
- which is epsilon close to optimal arm

PAC Bounds for MAB Naive Algorithm:

Input: $\epsilon > 0, \delta > 0$

Output: 'An arm'

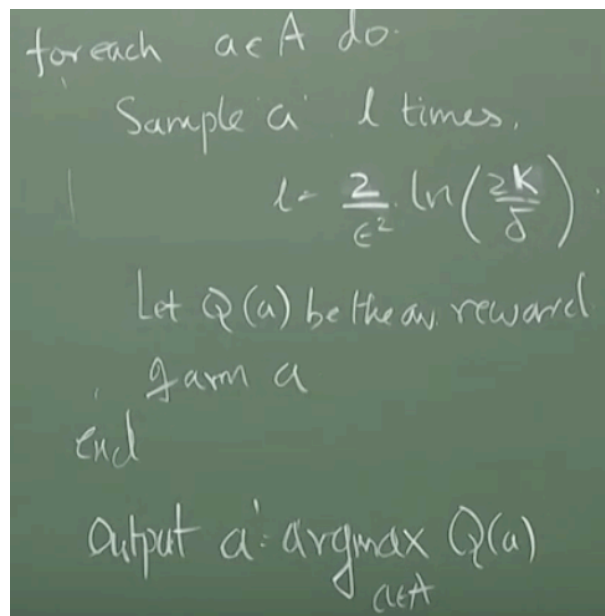
For each $a \in A$, do

Sample 'a' l times where $l = \frac{2}{\epsilon^2} \ln\left(\frac{2K}{\delta}\right)$

Let $Q(a)$ be the average reward of arm a

End

Output $a = \operatorname{argmax}_{a \in A} Q(a)$



foreach $a \in A$ do
 Sample 'a' l times,
 $l = \frac{2}{\epsilon^2} \ln\left(\frac{2K}{\delta}\right)$
 Let $Q(a)$ be the av. reward
 of arm a
end
 Output $a = \operatorname{argmax}_{a \in A} Q(a)$

I will take an arm, keep pulling until i get good estimate of expectation

- Will do for each arm
- Go and play the arm which has highest expectation

How did I produce the magic number?

Theorem 1: Naive (ϵ, δ) is an (ϵ, δ) - PAC algorithm
with arm sample complexity

$$O\left(\left(\frac{K}{\epsilon^2}\right) \log\left(\frac{K}{\delta}\right)\right).$$

for n arms do n time

Show that is actually (epsilon, delta) PAC

Let a' be an arm such that $q_*(a') < q_*(a^*) - \epsilon$.

$$P(Q(a') > Q(a^*)) \leq P(Q(a') > q_*(a') + \epsilon/2 \text{ OR } Q(a^*) < q_*(a^*) - \epsilon/2)$$

See explanation for why $\epsilon/2$ terms are added

$q^*(a^*)$ is optimal $\rightarrow$ pickup $q^*(a')$ - atleast close to epsilon

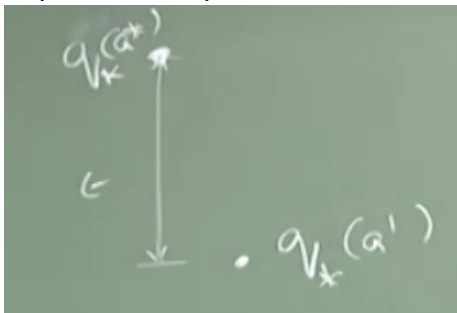
- Worry about making mistake - arm which is more that epsilon away

$$P(Q(a') > Q(a^*))$$

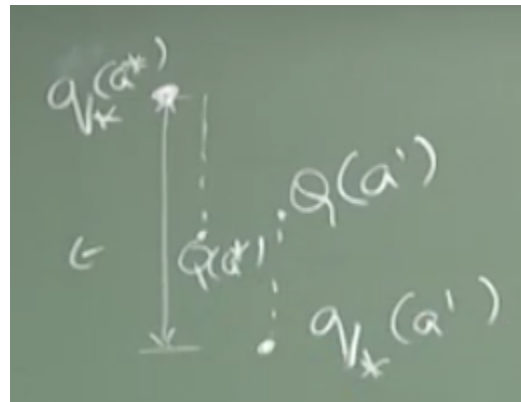
- To output a' this condition must be satisfied:
- It has to be better than a'
-

Look at its prob and bound it

Gap is atleast epsilon:



Making some estimation of $q^*(a^*)$ and $q_k(a')$



$Q(a^*)$ somewhat lower than $Q(q')$

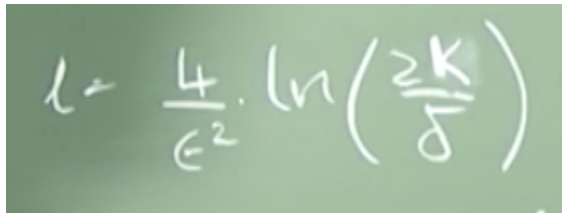
This means $Q(a')$ has to be more than $\epsilon/2$ above or below

- Think of case where $Q(a')$ will be underestimating or overestimating
- Then it is more than epsilon away

Taking them independent and adding

$$\begin{aligned} P(Q(a') > Q(a^*)) &\leq P(Q(a') > q_*(a') + \epsilon/2 \text{ OR } Q(a^*) < q_*(a^*) - \epsilon/2) \\ &\leq P(Q(a') > q_*(a') + \epsilon/2) + P(Q(a^*) < q_*(a^*) - \epsilon/2) \\ &\leq e^{-2\frac{\epsilon^2}{4}l} * 2 \end{aligned}$$

It is symmetrical so $+\epsilon/2$ or $-\epsilon/2 \rightarrow$ both are same so multiply by 2



Plug in $l = \frac{2}{\epsilon^2} \ln\left(\frac{2K}{\delta}\right)$

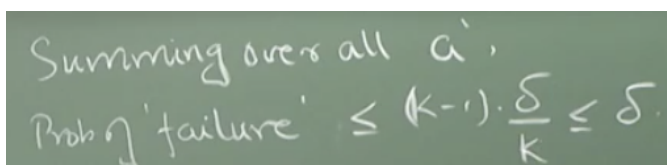
$$\begin{aligned} P(Q(a') > Q(a^*)) &\leq 2e^{-\frac{1}{2}\epsilon^2 \times \frac{2}{\epsilon^2} \ln\left(\frac{2K}{\delta}\right)} \\ &\leq 2e^{-\ln\left(\frac{2K}{\delta}\right)} \\ &\leq 2 \times \left(\frac{2K^{-1}}{\delta}\right) \\ &\leq \frac{\delta}{K} \end{aligned}$$

Prob of making mistake (for one a')
 = should it be δ - one non-optimal arm
 For all arms $\rightarrow$ Left with δ/K (or $K-1$, let us take K)

Now we know how we came up with magic number - derive so that we get δ/K

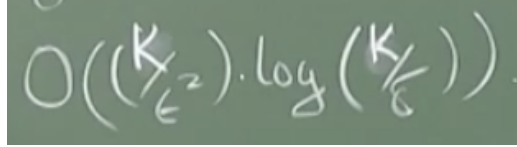
So that's just for 1 a' . Now summing over all a' ,

$$\text{Prob. Of failure} \leq (K-1) \frac{\delta}{K} \leq \delta$$



Analysis is pretty straightforward

- You can compute I


$$O\left(\left(\frac{K}{\epsilon^2}\right) \cdot \log\left(\frac{K}{\delta}\right)\right).$$

It is impossible to get rid of

This is sample complexity - K appears at two places

It is very hard to get rid of $1/\epsilon^2$ and $\ln(1/\delta)$

Can i get rid of K - it should depend atleast on one arm - so can't get rid of first K

Can try to get rid of K in $\ln(K/\delta)$

$$\frac{K}{\epsilon^2} \log\left(\frac{1}{\delta}\right)$$

You have to live with:

ToDo (from 18min): w2 L14 Median Elimination:

PAC Bounds for MAB Median Elimination Algorithm

PAC Bounds for MAB

Median Elimination Algo

Input: $\epsilon > 0, \delta > 0$

Output: An arm

Set $S_1 = A, \epsilon_1 = \epsilon/4, \delta_1 = \delta/2, l = 1$

Repeat

sample every arm $a \in S_l$ for

$\frac{1}{(\frac{\epsilon_l}{2})^2} \log\left(\frac{3}{\delta_l}\right)$. Let $Q_l(a)$ denote its value

Find the median of $Q_l(a)$, denoted by m_l

$S_{l+1} = S_l \setminus \{a: Q_l(a) < m_l\}$

$\epsilon_{l+1} = \frac{3}{4} \epsilon_l, \delta_{l+1} = \delta_l/2, l = l+1$

Until $|S_l| = 1$;

Input: $\epsilon > 0, \delta > 0$

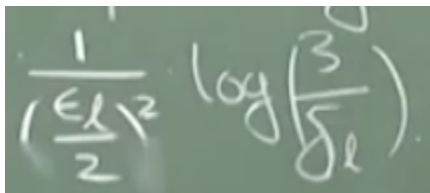
Output: An arm

Set $S_1 = A, \epsilon_1 = \epsilon/4, \delta_1 = \delta/2, l = 1$

Repeat

Sample every arm $a \in S_l$ for $\frac{1}{\epsilon_l^2/2} \log\left(\frac{3}{\delta_l}\right)$. Let $Q_l(a)$ denote its value.

Start off with K arms right and then in the first round okay
so median elimination is going to proceed in rounds okay.
In the first round we pull each arm some number of times right, determined by this magic
quantity.



Estimate Empirical value Q -fn forming at l -th stage - find median

Find the median of $Q_l(a)$, denoted by m_l .

$$S_{l+1} = S_l \setminus \{a: Q_l(a) < m_l\}$$

$$\epsilon_{l+1} = 3/4\epsilon_l; \delta_{l+1} = \delta_l/2; l = l + 1$$

Until $|S_l| = 1$;

Arrange q in ascending order. Eliminate all the arms whose value is less than median
 m_l is median at - current sl - $Q_l(a) < m_l$

Change constant by magic again - now becomes $3/4$, δ_{l+1} as $\delta_l/2$

- Keep going until no. of arms become 1

Will have $\log-k$ rounds

- Eliminating half arms

Sample complexity: $\frac{K}{\epsilon^2} \log\left(\frac{1}{\delta}\right)$

Show: I should not eliminate all epsimal optimal arms

- At every round doing is small
- So for all rounds it is below delta

Theorem:

Thm: The MEA(ϵ, δ) is a (ϵ, δ) -PAC
 Algo. w. sample complexity $O\left(\frac{K}{\epsilon^2} \cdot \log\left(\frac{1}{\delta}\right)\right)$

The MEA(ϵ, δ) is a (ϵ, δ) -PAC algorithm with sample complexity $O\left(\frac{K}{\epsilon^2} \log\left(\frac{1}{\delta}\right)\right)$

Sample complexity is trivial

$$\frac{1}{\left(\frac{\epsilon_l}{2}\right)^2} \log\left(\frac{3}{\delta_l}\right)$$

- Sum quantity for every arm (size of arm $K, K/2, \dots$)
- ϵ_l and δ_l will become smaller and smaller
- Sum over l (replace ϵ_l by fn of $\epsilon_{l-1}, \epsilon_{l-2}, \dots$)
- Write similar expression for δ_l

Lemma! For every phase l in MEA, we have.

$$\Pr\left[\max_{j \in S_l} q_{*}(j) \leq \max_{i \in S_{l+1}} q_{*}(i) + \epsilon_l\right] \geq 1 - \delta_l$$

Lemma: For every phase l in the MEA, we have the

$$\Pr\left[\max_{j \in S_l} q_{*}(j) \leq \max_{i \in S_{l+1}} q_{*}(i) + \epsilon_l\right] \geq 1 - \delta_l$$

$q_{*}(j)$ - true best arm at level l

At level 1 $\rightarrow q_{*}(a^{*})$ (not capital Q) $\rightarrow$ level 2 could be different as i might have lost a^{*} in first round itself

$$P \left[\max_{j \in S_l} q_{V_*}(j) \leq \right]$$

- Best i could have done in l-th round

$$P \left[\max_{j \in S_l} q_{V_*}(j) \leq \max_{i \in S_{l+1}} q_{V_*}(i) \right]$$

- Best i could have done in l+1-th round
- Bad case is when second term is smaller than first quantity

$$q_{V_*}(j) \leq q_{V_*}(i) + \epsilon$$

By how much should it be less - add epsilon - within epsilon
Acceptable

$$P \left[\max_{j \in S_l} q_{V_*}(j) \leq \max_{i \in S_{l+1}} q_{V_*}(i) + \epsilon_l \right] \geq 1 - \delta_l$$

Prob of good outcome is atleast 1-delta -> that means prob is high

It is ϵ -l and δ -l as every round i loose ϵ -l

- I shouldn't have lost more than ϵ
- Even if i lose ϵ -l -> overall loss is less than ϵ
- So will have atleast one optimal arm is left

Output: An arm

$$\text{Set } S_1 = A, \epsilon_1 = \epsilon/4, \delta_1 = \delta/2, l=1$$

That's the reason you start with less ϵ and δ

ToDo - 18min

W.log = without loss of generality

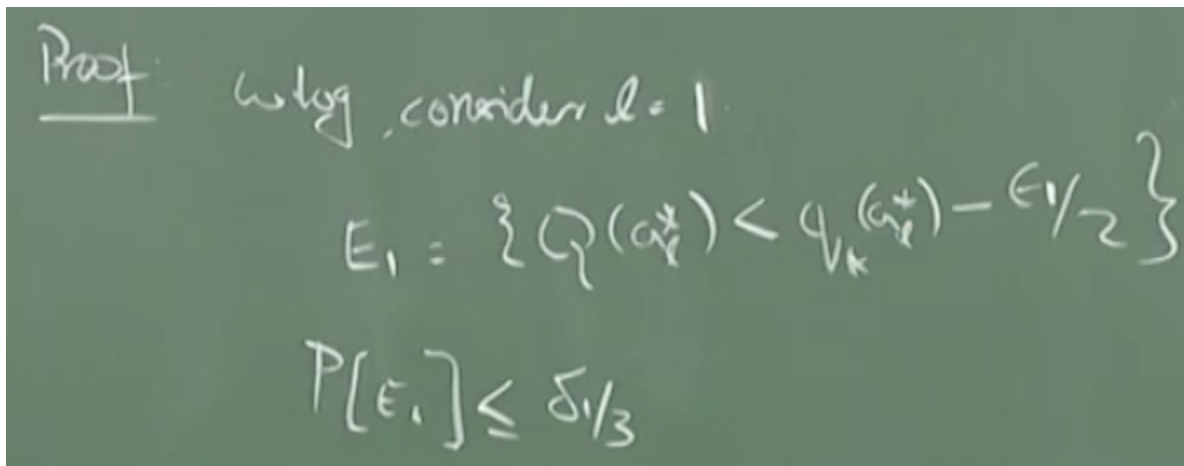
Lemma: For every phase l in the MEA, we have the

$$\Pr[\max_{j \in S_l} q_*(j) \leq \max_{i \in S_{l+1}} q_*(i) + \epsilon_l] \geq 1 - \delta_l$$

Proof:

W.log consider $l=1$,

$$E_1 = \{Q(a_l^*) < q_*(a_l^*) - \epsilon_1/2\}$$



Proof: W.log consider $l=1$.

$$E_1 = \{Q(a_l^*) < q_*(a_l^*) - \epsilon_1/2\}$$
$$P[E_1] \leq \delta_1/3$$

Look at first event E_1 : Grossly underestimating the true value

- True best arm in the l -th level

Apply Chernoff Bound, we get $P[E_1] \leq \delta_1/3$

This is the first event, suppose it doesn't hold. '

Consider event E_2 :

$$P[Q_l(j) \geq Q_l(a_l^*) | Q_l(a_l^*) > q_*(a_l^*) - \epsilon_l/2] ; j\text{th arm is not } \epsilon_l \text{ optimal}$$

$$P\left[Q_l(j) \geq Q_l(a_l^*) \mid Q_l(a_l^*) \geq q_*(a_l^*) - \epsilon_l/2\right]$$

j^{th} arm is not ϵ_l optimal.

Since we are conditioning on the event that $Q_l(a_l^*)$ is not under-estimated as in E1, therefore $Q_l(j)$ has to be over-estimated by more than $\epsilon/2$. Based on a similar reasoning we saw in the Naïve(ϵ, δ) PAC algorithm. So, we will get

$$P[Q_l(j) \geq Q_l(a_l^*) \mid Q_l(a_l^*) > q_*(a_l^*) - \epsilon_l/2] \leq \delta_1/3$$

Let $\#bad$ be the no. of bad arms such that the condition in E2 holds.

$$E[\#bad | Q_l(a_l^*) \geq q_*(a_l^*) - \epsilon_1/2] < |S_l| \frac{\delta_1}{3}$$

Now, to bound this,

$$P\left[\#bad \geq \frac{|S_l|}{2} \mid Q_l(a_l^*) \geq q_*(a_l^*) - \epsilon_1/2\right]$$

Apply Markov inequality,

$$P\left[\#bad \geq \frac{|S_l|}{2} \mid Q_l(a_l^*) \geq q_*(a_l^*) - \epsilon_1/2\right] \leq 2\delta_1/3$$

The total probability of bad events ie. E1 and E2 happening is therefore upper - bounded by δ_1
Therefore, the probability of the good outcome is

$$\Pr[\max_{j \in S_l} q_*(j) \leq \max_{i \in S_{l+1}} q_*(i) + \epsilon_l] \geq 1 - \delta_l$$

w2 L15 Thompson Sampling (13min):

Book - in passing mentions Median - we will build on these ideas when needed for RL

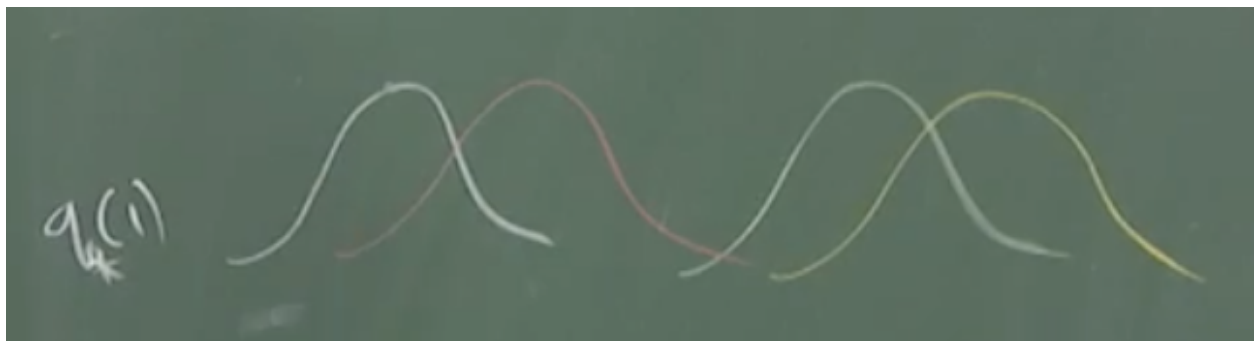
Bandit is taken as an Evil - Bandit is still - large area for Research

- Earlier did all of bandit in one class

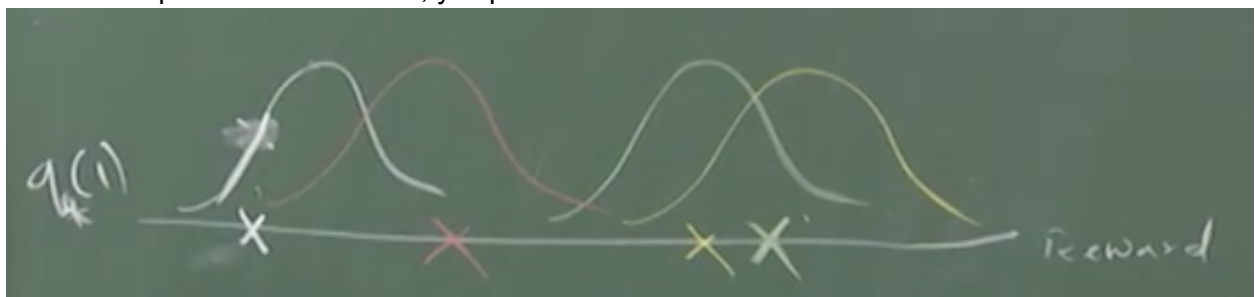
Gaining traction - called Posterior sampling

- Idea, if you know Q^* it is easy, take highest
- Take somewhat bayesian way of looking at the problem
- Will make assumptions about true q^* comes from some distribution
- If rewards are between 0 and 1
 - Proper assumption - Beta Distribution - limited to 0 and 1 and can
 - Take any kind of weird form - popular prior
 - I don't know anything - q^* is uniform - can come from any where
 - Random, all arms good/bad
 - Given current knowledge which arm should i play
 - Take each bandit problem/combination - q^*_1 q^*_2 etc, what is best arm
 - For each of bandit problem combination - what is the probability distribution
- Posterior
 - I have some distribution over q^* value
 - I draw sample from that distribution

$q^*_1, a^*_2, a^*_3, q^*_4$. Most probably q^* is somewhere



Draw a sample. X is reward axis, y is prob axis

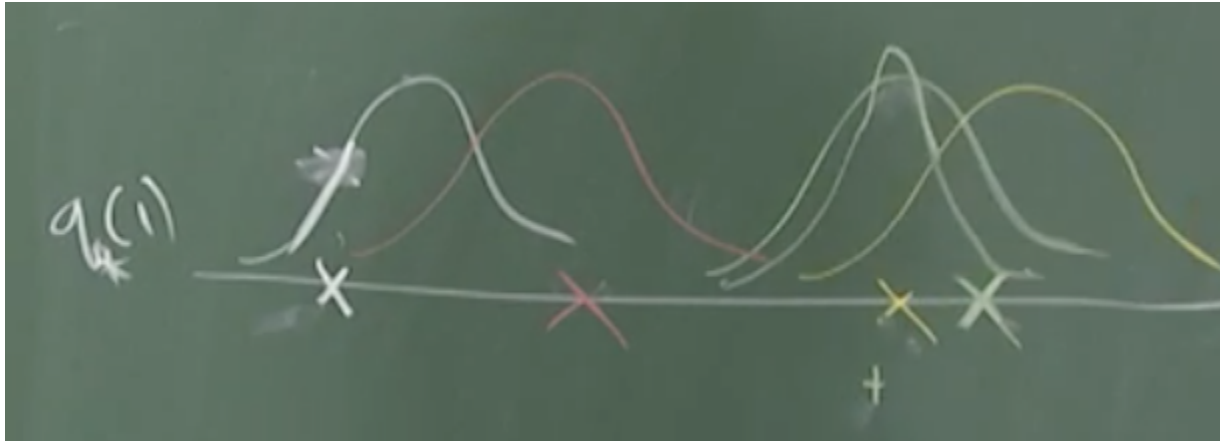


I have a specific bandit problem where my $q^*(1)$ is this $q^*(2)$ is this $q^*(3)$ is this and $q^*(4)$ is

- Drawn on bandit problem from this distribution

Pull arm-3 Look at actual reward get for arm-3

- Update this fellow ased on the reward



Shrink it or adjust it

It will become narrower because this is I kind of reinforcing my original belief that this is the most probable outcome.

- This is my belief
- Now will not sample from green
- It will converge to true problem - will be sampling from it

This can give better regret bounds than UCB

- Lot of attention paid to it
- Analysis is hairy - recently some papers - new ways of analyzing algorithms

Set of conditions:

- Showing guarantee

Very old algo

- Learning automata (finite state machines)
- Var structured automata
- Some of what they are doing is as posterior sampling doing
- They also satisfy regret bound $\ln n$

MEA:

Thm: The MEA(ϵ, δ) is a (ϵ, δ) -PAC

Algo. w. sample complexity $O\left(\frac{K}{\epsilon^2} \cdot \log\left(\frac{1}{\delta}\right)\right)$

Lemma! For every phase l in MEA, we have.

$$P\left[\max_{j \in S_l} q_{V_*}(j) \leq \max_{i \in S_{l+1}} q_{V_*}(i) + \epsilon_l\right] \geq 1 - \delta_l$$

Proof wlog, consider $l=1$.

$$E_1 = \{Q_1(a_1^*) < q_{V_*}(a_1^*) - \epsilon/2\}$$

$$P[E_1] \leq \delta/3$$

$$P\left[Q_1(j) \geq Q_1(a_1^*) \mid Q_1(a_1^*) \geq q_{V_*}(a_1^*) - \epsilon/2\right] \leq \delta/3 \quad - (1)$$

j^{th} arm is not ϵ_l optimal